

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Comparative Analysis
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	AI&ML/3 rd year	Date:	25/08/2026

Instructions to Students

- Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
- Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption.
- Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
- Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable.
- Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

QUESTIONS

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.
2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.
3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.
4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.
5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

Code of Conduct

I Kolar Umme Sadiya-192425117 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kolar Umme Sadiya

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Marks Evaluation:

Q. No	Scope of Items (20)	Comparison Depth (25)	Use of Evidence (20)	Critical Thinking (20)	Clarity of Presentation (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.

Cache memory is a small and high-speed memory located close to the CPU. It stores frequently used instructions and data so that the processor can access them faster than accessing main memory.

L1 Cache

L1 cache is the smallest and fastest cache memory. It is located closest to the CPU core and normally contains separate instruction and data caches. It has the lowest access latency and provides very high bandwidth. Because of its small size, it stores only the most frequently accessed data and instructions.

L2 Cache

L2 cache is larger than L1 but slightly slower. When the required data is not found in L1, the processor checks L2. It provides a balance between capacity and speed. It reduces the number of accesses to slower main memory and improves system throughput.

L3 Cache

L3 cache is larger than L1 and L2 but has higher access latency. It is commonly shared among multiple processor cores. It stores frequently shared data and reduces the need to access DRAM.

Comparison

Feature	L1 Cache	L2 Cache	L3 Cache
Size	Smallest	Larger than L1	Largest
Access Latency	Lowest	Low	Higher
Speed	Fastest	Very Fast	Fast
Bandwidth	Highest	High	Lower than L1/L2
Capacity	Small	Medium	Large
Usage	Frequently used data	Backup for L1	Shared data

Influence on Processor Performance

Cache memory reduces the time required by the CPU to access frequently used data. If data is found in the cache, it is called a cache hit. A cache miss forces the processor to access the next cache level or main memory, increasing latency.

L1 provides the fastest access, L2 provides additional capacity with slightly higher latency, and L3 reduces expensive accesses to main memory. Therefore, a well-designed cache hierarchy increases processor performance, improves memory access efficiency, and reduces CPU waiting time.

Conclusion

L1, L2, and L3 caches work together to reduce memory access latency. L1 provides the fastest access, L2 provides a balance between speed and capacity, and L3 provides larger shared storage. A high cache hit rate improves processor performance and overall system throughput.

2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.

SRAM and DRAM are two types of volatile memory. Both are used in computer systems, but their characteristics are different.

SRAM

SRAM stands for Static Random Access Memory. It provides very low access latency and high-speed operation. SRAM does not require periodic refreshing as long as power is supplied. However, it requires more transistors per bit, making it more expensive and less dense than DRAM.

DRAM

DRAM stands for Dynamic Random Access Memory. It stores data using capacitors and requires periodic refreshing. It has higher access latency than SRAM but provides higher density and larger capacity at a lower cost per bit.

Comparison

Feature	SRAM	DRAM
Access Latency	Very Low	Higher
Speed	Very High	High
Bandwidth	High	High
Density	Low	High
Capacity	Smaller	Larger
Cost per Bit	High	Lower
Refresh	Not Required	Required
Typical Use	Cache Memory	Main Memory

Why SRAM is Used in Cache Memory

Cache memory must provide extremely fast access because it directly affects CPU performance. SRAM has very low latency and high speed, making it suitable for L1, L2, and L3 cache memory.

Why DRAM is Used in Main Memory

Main memory requires much larger capacity than cache memory. DRAM provides high density and lower cost per bit, making it possible to build large-capacity main memory economically.

Conclusion

SRAM is preferred for cache memory because of its low latency and high speed. DRAM is preferred for main memory because of its high density, large capacity, and lower cost. Thus, each technology is used according to its performance characteristics.

3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.

Virtual memory, paging, and cache memory are important mechanisms used to manage memory efficiently and improve computer system performance.

Cache Memory

Cache memory stores frequently accessed instructions and data close to the CPU. When the requested data is present in the cache, it is called a cache hit. The CPU can access the data quickly, resulting in very low latency.

If the required data is not present, it is called a cache miss. The processor must access a lower level of memory, increasing access time.

Paging

Paging divides virtual memory into fixed-size blocks called pages and physical memory into blocks called frames. Pages can be loaded into available physical frames when required.

Paging improves memory utilization because physical memory can be allocated efficiently among multiple programs.

Virtual Memory

Virtual memory provides each process with a virtual address space. It allows programs to use more logical memory than the available physical RAM by using secondary storage when necessary.

However, accessing secondary storage is much slower than accessing RAM. Therefore, frequent page faults can increase latency and reduce system throughput.

Comparison

Mechanism	Main Purpose	Access Latency	Effect on Memory Utilization	Effect on Performance
Cache	Fast data access	Very Low	Efficient use of fast memory	Improves CPU performance
Paging	Memory allocation	Moderate	Improves physical memory utilization	Supports multitasking
Virtual Memory	Provides larger logical memory	High when storage is accessed	Allows larger programs	Can reduce performance during frequent page faults

Contribution to System Performance

Cache memory reduces CPU waiting time by providing frequently used data quickly. Paging allows several processes to use physical memory efficiently. Virtual memory allows programs that require more memory than available RAM to execute.

However, excessive page faults can cause a condition called thrashing, where the system spends more time transferring pages than executing programs. This reduces throughput.

Conclusion

Cache memory improves access speed, paging improves memory allocation, and virtual memory provides flexibility and supports large programs. Together, they contribute to efficient memory management and overall system performance.

4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.

NAND Flash and DRAM are important memory technologies, but they are designed for different purposes in the memory hierarchy.

DRAM

DRAM is volatile memory used mainly as the computer's main memory. It provides relatively low access latency and high bandwidth compared with secondary storage.

DRAM allows the CPU to quickly access active programs and data. However, it loses stored information when power is removed.

NAND Flash

NAND Flash is a non-volatile memory technology. It retains data even when the power is switched off. It provides high storage capacity at a lower cost per bit than DRAM.

NAND Flash is commonly used in SSDs and other secondary storage devices. Modern NAND-based SSDs can provide high sequential throughput, but their individual access latency is still higher than DRAM.

Comparison

Feature	DRAM	NAND Flash
Type	Volatile	Non-Volatile
Access Latency	Low	Higher
Bandwidth	High	High for block/sequential transfers
Capacity	Large	Very Large
Cost per Bit	Higher	Lower
Data Retention	Requires power	Retains data without power
Main Application	Main Memory	Secondary Storage

Suitability in Memory Hierarchy

DRAM is suitable for main memory because the processor frequently needs fast access to active instructions and data.

NAND Flash is suitable for secondary storage because it provides large capacity, non-volatility, and lower cost per bit.

Neither DRAM nor NAND Flash is normally used as the primary CPU cache because CPU cache requires extremely low latency.

Conclusion

DRAM provides fast access and is suitable for main memory, while NAND Flash provides large, persistent storage and is suitable for secondary storage. Their different characteristics allow them to occupy different levels of the hierarchical memory system.

5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

MRAM and RRAM are emerging non-volatile memory technologies. They have the potential to improve future memory systems by combining persistence with relatively fast access.

MRAM

MRAM stands for Magnetoresistive Random Access Memory. It stores information using magnetic states. It is non-volatile, meaning that data is retained even when power is removed.

MRAM can provide fast access, good endurance, and low standby power. These characteristics make it suitable for embedded systems and future high-performance memory applications.

Advantages of MRAM

- Non-volatile operation
- Fast access
- Good endurance
- Low standby power
- Data is retained without continuous power
- Potential for embedded and high-performance applications

Limitations of MRAM

- Higher manufacturing complexity
- Higher cost compared with conventional memory
- Integration challenges with semiconductor manufacturing

RRAM

RRAM stands for Resistive Random Access Memory. It stores information by changing the resistance of a material. It is also non-volatile and has the potential for high density and low energy consumption.

Advantages of RRAM

- Non-volatile operation
- High-density potential
- Low-power operation
- Fast switching potential
- Potential use in future storage-class memory
- Suitable for embedded applications

Limitations of RRAM

- Device-to-device variability
- Endurance challenges depending on implementation
- Manufacturing and reliability challenges
- Less mature than conventional DRAM and NAND Flash

Comparison

Feature	MRAM	RRAM
Volatility	Non-Volatile	Non-Volatile
Access Time	Fast	Potentially Fast
Throughput	High Potential	High Potential
Energy Efficiency	Good	Potentially Very Good
Endurance	Generally High	Technology Dependent
Main Advantage	Speed and Endurance	Density and Low-Power Potential
Main Limitation	Cost and Integration	Variability and Reliability

Future Applications

MRAM can be used in embedded memory, fast non-volatile memory, and systems where speed and data persistence are important.

RRAM has potential applications in high-density non-volatile memory, storage-class memory, and energy-efficient computing systems.

Both technologies could help reduce the gap between conventional volatile memory and persistent storage.

Conclusion

MRAM provides a strong combination of fast access, endurance, and non-volatility. RRAM provides strong potential for high density, fast switching, and energy efficiency. However, manufacturing, reliability, cost, and endurance challenges must be addressed before they can widely replace conventional memory technologies.

Overall Conclusion

A computer system uses a hierarchical memory system because no single memory technology can provide the lowest latency, highest capacity, lowest cost, and non-volatility at the same time.

The memory hierarchy can be represented as:

L1 Cache → L2 Cache → L3 Cache → DRAM → NAND Flash

As we move down the hierarchy:

- Capacity generally increases.
- Cost per bit generally decreases.
- Access latency generally increases.
- Speed generally decreases.
- Persistent storage becomes more important.

SRAM is suitable for cache because of its low latency and high speed. DRAM is suitable for main memory because of its high density and capacity. NAND Flash is suitable for secondary storage because of its large capacity and non-volatile nature. Virtual memory and paging improve memory utilization, while cache memory reduces effective memory access latency.

MRAM and RRAM are promising technologies for future hierarchical memory systems because they provide non-volatility with the potential for fast access, high throughput, and improved energy efficiency. Therefore, an efficient hierarchical memory system combines different technologies based on **latency, bandwidth, throughput, capacity, energy efficiency, cost, and endurance**. This combination helps reduce memory bottlenecks and improves overall computer system performance.